

CURRICULUM VITAE

Yue Chu, PhD, MSPH, MA, MBBS

PERSONAL DATA

1885 Neil Avenue Mall
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Department of Sociology
The Ohio State University
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EDUCATION AND TRAINING

PhD/2025 The Ohio State University, Columbus, USA
Major: Sociology (Advisor: Dr. Samuel Clark)
Minor: Statistics, Graduate Interdisciplinary Specialization in Demography
Dissertation: Leveraging Language Models and Machine Learning in Verbal Autopsy Analysis

MA/2020 The Ohio State University, Columbus, USA
Major: Sociology (Advisor: Dr. Samuel Clark)
Thesis: SVD-Bayes: A Singular Value Decomposition-Based Approach Under Bayesian Framework for Indirect Estimation of Age-Specific Fertility and Mortality

MSPH/2014 Johns Hopkins Bloomberg School of Public Health, Baltimore, USA
Concentration: Population, Family and Reproductive Health

MBBS/2012 Fudan University, Shanghai, China
Specialty: Preventive Medicine

PROFESSIONAL EXPERIENCE

May 2025 – present	Post-doctoral fellow, Health and Environment Modeling Laboratory, the Ohio State University, Columbus, OH
May 2023 – present	Consultant on Reference Death Archive project for World Health Organization
Aug 2023 – Sep 2023	Visiting scholar at Africa Health Research Institute, Somkele, South Africa
May 2022 – May 2023	Data Science intern, Nationwide Mutual Insurance Company / Nationwide Center for Advanced Customer Insights, Columbus, OH
Dec 2022 – Jan 2023	Consultant on Survey ODK for Ministry of Health, Guinea-Bissau
Apr 2021 – May 2021	Consultant on WHO VARG Workshop for WHO verbal autopsy instrument revision

Jan 2019 – May 2025	Graduate Research Associate, Institute for Population Research, the Ohio State University, Columbus, OH
Jul 2018 – Aug 2018	Consultant, Maternal, Newborn, Child and Adolescent Health Division, World Health Organization, Geneva, Switzerland
Jul 2018 – Aug 2018	Consultant, KIT Health Royal Tropical Institute, Amsterdam, Netherlands
Jan 2015 – Jul 2018	Research Associate, Department of International Health, Johns Hopkins Bloomberg School of Public Health
Sep 2014 – Jan 2015	Research Assistant, Department of Health Policy and Management, Johns Hopkins Bloomberg School of Public Health
Apr 2013 – Sep 2014	Research assistant, Johns Hopkins Bloomberg School of Public Health
May 2013 – Aug 2013	Research Assistant, Technical Division, United Nations Population Fund (UNFPA), New York
Apr 2012 – Jun 2012	Research Assistant, Department of Health Statistics and Social Medicine, School of Public Health, Fudan University, Shanghai, China
Feb 2012 – Mar 2012	Intern, Division of Vital Statistics, Center of Disease Control and Prevention, Shanghai, China
Mar 2011 – Aug 2011	Clinical Internship, Fifth People's Hospital, Shanghai, China
Dec 2009 – Apr 2011	Student principal investigator, Department of Maternal and Child Health, Fudan University, Shanghai, China

PROFESSIONAL DEVELOPMENT

2025	PAA workshop: A Primer on Machine Learning and Artificial Intelligence for Population Research
2024	Max Planck Institute for Demographic Research workshop: Demystifying Machine Learning
2022	Berkeley Population Center: Annual Workshop on Formal Demography
2021	Pedagogical training with The Drake Institute for Teaching and Learning, Columbus, USA
2019	Data Carpentry workshop in Social Sciences, Columbus, USA
2018	D4H/WHO workshop on Verbal Autopsy Methods, Columbus, USA
2018	IUSSP workshop: Bayesian Small Area Estimation using Complex Survey Data: Methods and Applications, Denver, USA
2018	Survey Research Design, Johns Hopkins Bloomberg School of Public Health, Baltimore, USA
2017	PAA workshop: Social Media and Demographic Methods, Chicago, USA
2017	Methods and Measures in Population Studies, Johns Hopkins Bloomberg School of Public Health, Baltimore, USA
2016	PAA workshop: Bayesian Population Projections: Theory and Practice, Washington DC, USA

2015 Analysis of Longitudinal Data, Johns Hopkins Bloomberg School of Public Health, Baltimore, USA

PROFESSIONAL ACTIVITIES

Society Membership

Population Association of America (PAA): 2015-present

International Union for the Scientific Study of Population (IUSSP): 2019-present

Center Affiliation

Health and Environment Modeling Laboratory, the Ohio State University: 2024-present

Institute for Population Research, the Ohio State University: 2018-present

Translational Data Analytics Institute, the Ohio State University: 2018-present

Nationwide Center for Advanced Customer Insights, the Ohio State University: 2022-2023

Johns Hopkins Institute for International Programs: 2015-2018

Johns Hopkins Evidence-based Practice Center: 2014

Johns Hopkins Center for Adolescent Health: 2013-2014

Conference organization

Session chair & discussant of Annual Meeting of Population Association of America.(2025)

Session chair of Annual Meeting of Population Association of America.(2021)

Session chair of Annual Meeting of Applied Linguistics and Professional Practice.(2021)

The Sixth Child Survival Forum in China, September 2016, Chengdu, China

Technical Advisory Group Annual Meeting on High Impact Practices in Family Planning, UNFPA, June 2013, New York, USA

EDITORIAL ACTIVITIES

Peer Review Activities

PNAS, Demographic Research, Population Health Metrics, Annals of Epidemiology, Nature, Journal of Global Health, Preventing Chronic Disease, European Journal of Population, PLOS One, PLOS Global Health, Clinical epidemiology, BMC Public Health, BMC Pediatrics, Journal of Epidemiology and Global Health, Frontiers, Population, Clinical Epidemiology and Global Health, Journal of Human Anatomy

HONORS AND AWARDS

Distinguished University Fellowship, The Ohio State University (2018,2023)

Tuition Scholarship, Johns Hopkins Bloomberg School of Public Health (2013)

People's Scholarship of Fudan University (2007-2009,2011-2012)

PUBLICATIONS

Peer-Reviewed Journal Articles

* indicate joint first authorship

1. **Chu Y.** *Leveraging language models and machine learning in verbal autopsy analysis* [preprint]. arXiv:2508.19274 [cs.CL]. 2025 Aug 22 [cited 2025 Aug 27]. Available from: <https://doi.org/10.48550/arXiv.2508.19274>
2. **Chu Y**, Marston M, Dube A, Festo C, Geubbels E, Gregson S, Herbst K, et al. Temporal changes in cause of death among adolescents and adults in six countries in eastern and southern Africa in 1995–2019: a multi-country surveillance study of verbal autopsy data. *Lancet Glob Health*. 2024;12(8):e1278–87.
3. Damerow SM, Yeung D, Martins JS, Pathak I, **Chu Y**, Liu L, Fisker AB. Provider-mother interactions are associated with birth outcome misclassifications in household surveys: A case-control study in Guinea-Bissau. *J Glob Health*. 2023;13:01005.
4. Liu Y*, **Chu Y***, Yeung D, Wang W, Wang L, Yin P, Liu J, Zhou M, Liu L. National and subnational levels and causes of mortality among 5–19-year olds in China in 2004–2019: Empirical evidence from the Disease Surveillance Points System. *J Glob Health*. 2022;12:04047.
5. Perin J, **Chu Y**, Villaviciencio F, Schumacher A, McCormick T, Guillot M, Liu L. Adapting and validating the log quadratic model to derive under-five age- and cause-specific mortality (U5ACSM): A preliminary analysis. *Popul Health Metr*. 2022;20(1):1–12.
6. Turner AN, Kline D, Norris A, Phillips WG, Root E, Wakefield J, Li Z, et al. Prevalence of current and past COVID-19 in Ohio adults. *Ann Epidemiol*. 2022;67(1):50–60.
7. Schumacher AE, McCormick TH, Wakefield J, **Chu Y**, Perin J, Villaviciencio F, Simon N, Liu L. A flexible Bayesian framework to estimate age- and cause-specific child mortality over time from sample registration data. *Ann Appl Stat*. 2022;16(1):124–43.
8. Kline D, Li Z, **Chu Y**, Wakefield J, Miller WC, Turner AN, Clark SJ. Estimating seroprevalence of SARS-CoV-2 in Ohio: A Bayesian multilevel poststratification approach with multiple diagnostic tests. *Proc Natl Acad Sci U S A*. 2021;118(26):e2029042118.
9. **Chu Y.** SVD-Bayes: A singular value decomposition-based approach under Bayesian framework for indirect estimation of age-specific fertility and mortality [master's thesis]. The Ohio State University; 2020. Available from: https://rave.ohiolink.edu/etdc/view?acc_num=osu1609638415015896
10. Helleringer S, Liu L, **Chu Y**, Rodrigues A, Fisker AB. Biases in survey estimates of neonatal mortality: Results from a validation study in urban areas of Guinea-Bissau. *Demography*. 2020;57(5):1705–26.
11. Wahl B, Deloria Knoll M, Shet A, Gupta M, Kumar R, Liu L, **Chu Y**, et al. National, regional, and state-level pneumonia and severe pneumonia morbidity in children in India: Modelled estimates for 2000 and 2015. *Lancet Child Adolesc Health*. 2020;4(9):678–87.
12. Liu L*, **Chu Y***, Oza S, Hogan D, Perin J, Bassani DG, Ram U, et al. National, regional, and state-level all-cause and cause-specific under-5 mortality in India in 2000–15: A systematic analysis with implications for the Sustainable Development Goals. *Lancet Glob Health*. 2019;7(6):e721–34.
13. Wahl B, Sharan A, Deloria Knoll M, Kumar R, Liu L, **Chu Y**, McAllister DA, et al. National, regional, and state-level burden of *Streptococcus pneumoniae* and *Haemophilus*

- influenzae* type b disease in children in India: Modelled estimates for 2000–15. *Lancet Glob Health*. 2019;7(6):e735–47.
14. Choudhary TS, Sinha B, Khera A, Bhandari N, **Chu Y**, Jackson B, Walker N, et al. Factors associated with the decline in under-five diarrhea mortality in India: A LiST analysis. *J Glob Health*. 2019;9(2):020418.
 15. Black R, Fontaine O, Lamberti L, Bhan M, Huicho L, El Arifeen S, Masanja H, et al. Drivers of the reduction in childhood diarrhea mortality 1980–2015 and interventions to eliminate preventable diarrhea deaths by 2030. *J Glob Health*. 2019;9(2):020801.
 16. Countdown to 2030 Collaboration. Countdown to 2030: Tracking progress towards universal coverage for reproductive, maternal, newborn and child health. *Lancet*. 2018;391(10129):1538–48.
 17. McAllister DA, Liu L, Shi T, **Chu Y**, Reed C, Burrows J, Adedoye D, et al. Global, regional, and national estimates of pneumonia morbidity and mortality in children younger than 5 years between 2000 and 2015: A systematic analysis. *Lancet Glob Health*. 2019;7(1):e47–57.
 18. Wahl B, O'Brien KL, Greenbaum A, Majumder A, Liu L, **Chu Y**, Lukšić I, et al. Global, regional, and national burden of *Streptococcus pneumoniae* and *Haemophilus influenzae* type b in children in the era of conjugate vaccines: Updated estimates from 2000–2015. *Lancet Glob Health*. 2018;6(7):e744–57.
 19. Wilson LM, Rebholz CM, Jirru E, Liu MC, Zhang A, Gayleard J, **Chu Y**, Robinson KA. Benefits and harms of osteoporosis medications in patients with chronic kidney disease: A systematic review and meta-analysis. *Ann Intern Med*. 2017;166(9):649–58.
 20. Liu L, Oza S, Hogan D, **Chu Y**, Perin J, Zhu J, Lawn JE, et al. Global, regional and national causes of child mortality in 2000–2015: Implications for the Sustainable Development Goals. *Lancet*. 2016 Nov 10. pii: S0140-6736(16)31593-8.
 21. He C, Liu L, **Chu Y**, Perin J, Dai L, Li X, Miao L, et al. National and subnational all-cause and cause-specific child mortality in China, 1996–2015: A systematic analysis with implications for the Sustainable Development Goals. *Lancet Glob Health*. 2017;5(2):e186–97.
 22. Liu L, Kalter HD, **Chu Y**, Kazmi N, Koffi AK, Amouzou A, Joos O, Munos M, Black RE. Understanding misclassification between neonatal deaths and stillbirths: Empirical evidence from Malawi. *PLoS One*. 2016;11(12):e0168743.
 23. Song P, Theodoratou E, Li X, Liu L, **Chu Y**, Black RE, Campbell H, Rudan I, Chan KY. Causes of death in children younger than five years in China in 2015: An updated systematic analysis. *J Glob Health*. 2016;6(2):020802.
 24. Maruthur NM, Tseng E, Hutfless S, Wilson LM, Liu MC, Zhang A, **Chu Y**, et al. Diabetes medications as monotherapy or metformin-based combination therapy for type 2 diabetes. *Ann Intern Med*. 2016;164(11):740–51.
 25. Bolen S, Tseng E, Hutfless S, Segal JB, Suarez-Cuervo C, Berger Z, Wilson LM, **Chu Y**, et al. Medications for adults with type 2 diabetes: An update focused on monotherapy and add-on therapy to metformin (EHC-2014-11-0075.R4). *AHRQ Comparative Effectiveness Review Report No. 16-EHC013-EF*. Rockville, MD, USA. 2015. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK362863/>

Book Chapters

Li Liu, Kenneth Hill, Shefali Oza, Dan Hogan, **Yue Chu**, Simon Cousens, Colin Mathers, Cynthia Stanton, Joy Lawn, Robert E. Black. 2016. "Chapter Four: Levels and Causes of Mortality under Age Five". *Reproductive, Maternal, Newborn and Child Health*. Disease Control Priorities, Third Edition, Vol2, Part 1. The World Bank. Washington DC.

CURRICULUM VITAE

Yue Chu

Part II

TEACHING

Teaching

SOC 3549 (OSU)	Recitation leader for Statistics in Sociology. 2021
380.603 (JHSPH)	TA for Demographic Methods in Public Health. 2013-2014
380.651 (JHSPH)	TA for Methods and Measures in Population Studies. 2016-2017

Advisee

Elenzie France. Undergraduate RA for SP24, The Ohio State University
Ambre Flowers. Undergraduate. JHSPH 2018 Diversity Summer Internship Program
Jayla Johnson. Undergraduate. JHSPH 2017 Diversity Summer Internship Program

Workshops

SGSA VP of research committee workshops on R (2022)
Verbal Autopsy data analysis workshops (2018, 2019, 2020)

PRESENTATIONS

Presentation

*indicates role as presenter

1. **Chu Y,*** Clark S. "Automating cause of death classification from verbal autopsy: Using pretrained language models and multimodal learning." *2025 Annual Meeting of the Population Association of America*, Columbus, OH. Apr 13, 2025. (oral)
2. **Chu Y.*** "Automating cause of death classification." *HEALMOD Professional Seminar*, Columbus, OH. Apr 1, 2025. (oral)
3. **Chu Y,*** Li Z, Wu Z, Clark S. "Leveraging machine learning techniques for verbal autopsy analysis." *Demystifying Machine Learning Workshop*, Max Planck Institute for Demographic Research, Rostock, Germany. Nov 5, 2024. (oral)
4. **Chu Y,*** Marston M, Dube A, Festo C, Geubbels E, Gregson S, Herbst K, et al. "Temporal changes in cause of death among adults in six countries in Eastern and Southern Africa: A multi-country cohort study using verbal autopsy data." *2023 Annual Meeting of the Population Association of America*, New Orleans, LA. Apr 14, 2023. (oral)
5. **Chu Y,*** Clark S. "Indirect estimation of age-specific fertility and mortality: SVD-Bayes model." *TDAI Fall Forum*, The Ohio State University, Columbus, OH. 2021. (oral)

6. **Chu Y,*** Clark S. "Indirect estimation of age-specific fertility and mortality: SVD-Bayes model." *2021 Annual Meeting of the Population Association of America*, St. Louis, MO. May 9, 2021. (oral)
7. Hayford S, Casterline J, **Chu Y.** "Changes in age at last birth over the course of contemporary fertility declines." *2021 Annual Meeting of the Population Association of America*, St. Louis, MO. May 9, 2021. (oral)
8. **Chu Y,*** Clark S. "Indirect estimation of age-specific fertility and mortality: SVD-Bayes model." *35th Annual Hayes Graduate Research Forum*, Columbus, OH. Apr 9, 2021. (poster)
9. Liu L, **Chu Y**, Rodrigues A, Fisker A, Helleringer S. "Understanding reporting errors of population survey-based neonatal mortality: A validation study from Guinea-Bissau." *2019 Annual Meeting of the Population Association of America*, Austin, TX. Apr 10–13, 2019. (oral)
10. **Chu Y,*** Yeung D, Liu L, Black R. "Global estimates for meningitis-specific mortality." *Meningitis Disease Burden Evaluation Alignment Meeting of WHO/Bill and Melinda Gates Foundation*, Seattle, WA. Nov 6, 2018. (oral)
11. Perin J, **Chu Y**, Schumacher A, McCormick T, Liu L. "Cause-specific mortality rates for fine age groups among children under five." *2018 Annual Meeting of the Population Association of America*, Denver, CO. Apr 26–28, 2018. (oral)
12. Liu L, **Chu Y.** "China Maternal and Child Health Surveillance System." *Global Age Patterns of Under-Five Mortality Project Kickoff Meeting*, Philadelphia, PA. Dec 18, 2017. (oral)
13. Perin J, Liu L, **Chu Y.** "Age-specific child causes of death." *IUSSP International Seminar on Mortality: Past, Present and Future*, Campinas, Brazil. Aug 7–8, 2017. (oral)
14. Liu L, Kalter H, Chu Y, et al. "Understanding misclassification between neonatal deaths and stillbirths: Empirical evidence from Malawi." *2017 Annual Meeting of the Population Association of America*, Chicago, IL. Apr 27–29, 2017. (oral)
15. He C, Liu L, **Chu Y,*** et al. "National and subnational levels and causes of child mortality in China, 1996–2015: Implications for the SDGs." *2017 Annual Meeting of the Population Association of America*, Chicago, IL. Apr 27–29, 2017. (oral)
16. He C, Liu L, **Chu Y,*** et al. "Estimating national and subnational causes of child deaths using data from the China Maternal and Child Health Surveillance System." *6th China Child Survival Forum*, Chengdu, China. Sep 8–9, 2016. (oral)
17. Liu L, Kalter H, **Chu Y,*** et al. "Understanding misclassification between neonatal deaths and stillbirths: Empirical evidence from Malawi." *WHO Informal Technical Consultation on Maternal and Neonatal Deaths and Stillbirths*, Geneva, Switzerland. Aug 9–11, 2016. (oral)
18. He C, Liu L, **Chu Y,*** et al. "Estimating national and subnational causes of child deaths using data from the China Maternal and Child Health Surveillance System." *WHO Informal Technical Consultation on Maternal and Neonatal Deaths and Stillbirths*, Geneva, Switzerland. Aug 9–11, 2016. (oral)
19. Wahl B, O'Brien K, Greenbaum A, et al. "Global burden of *Streptococcus pneumoniae* in children under 5 in the era of PCV: 2000–2015." *10th International Symposium on Pneumococci and Pneumococcal Disease*, Glasgow, Scotland. Jun 27–30, 2016. (oral)

20. Wahl B, Arora N, Kapoor AN, et al. "Mortality caused by *Streptococcus pneumoniae* in children under 5 in India: A state-by-state analysis from 2000 to 2015." *10th International Symposium on Pneumococci and Pneumococcal Disease*, Glasgow, Scotland. Jun 27–30, 2016. (oral)
21. Liu L, Oza S, Hogan D, **Chu Y**, et al. "Global, regional, and national causes of child mortality in 2000–2015: Reflecting on MDG 4 and prioritizing for SDG 3.2." *2016 Annual Meeting of the Population Association of America*, Washington, DC. Apr 2, 2016. (oral)
22. Liu L, **Chu Y**, Black R. "Subnational child cause of death estimation in China." *UN IGME Technical Advisory Group Bi-annual Meeting*, Washington, DC. Dec 9, 2015. (oral)
23. Black R, Liu L, **Chu Y**. "Causes of child mortality in 2000–2015: The global picture and current state in Nigeria." *Nigeria Verbal/Social Autopsy Technical Working Group Meeting*, Abuja, Nigeria. Nov 23, 2015. (oral)
24. Liu L, Oza S, **Chu Y**, et al. "Progress in estimating causes of child mortality in 2000–2015." *UN IGME Technical Advisory Group Bi-annual Meeting*, New York, NY. May 26, 2015. (oral)
25. **Chu Y**,* Liu L, Black R. "Progress in estimating causes of death among children aged 1–59 months – Update for 2000–2015." *Maternal and Child Epidemiology Estimation Project Review Meeting*, Geneva, Switzerland. Apr 20–21, 2015. (oral)
26. **Chu Y**,* Liu L. "Differences in cause-of-death distribution estimates in China: MCMSS vs MCEE." *Maternal and Child Epidemiology Estimation Project Review Meeting*, Geneva, Switzerland. Apr 20–21, 2015. (oral)

SKILLS

Statistical software: R, Python, Julia, Stata, SAS, SPSS, SQLite/SQL, RJAGS, RNIMBLE, JMP.
Data science toolkits: Hugging Face, Optuna, Prompt engineering, Azure, Kubeflow, Snowflake, Teradata, DBeaver, Amazon Web Services (AWS/S3).

ADDITIONAL INFORMATION

I am a population health expert, demographer, epidemiologist, and data scientist, with a multidisciplinary background spanning global health, demography, epidemiology, sociology, applied statistics, and clinical medicine.

My current research focuses on applying advanced statistical and demographic methods to address key questions in population health estimation, particularly in resource-limited settings. My work bridges cutting-edge technologies and methodologies — such as language models, multimodal fusion, data science, Bayesian models — with most challenging public health issues in some of the world's most pressing and underserved areas, ensuring practical impact and real-world relevance.

Currently I'm involved in various projects including:

- Leveraging language models and machine learning to enhance data collection and analysis in low- and middle-income countries.
- Developing the Reference Data Archive, a centralized repository for survey and surveillance data with a publicly accessible catalog and trusted research environment.
- Assessing and improving data quality in population surveys and demographic surveillance systems.
- Indirect estimation of mortality and fertility using Bayesian models.

Key words: demography, global health, population health, verbal autopsy, mortality, vital statistics, demographic surveillance, survey methodology, applied statistics, Bayesian modeling, data science, natural language processing, computational social science, data engineering.